

DATA PREPROCESSING USING PYTHON

Machine Learning / Data Science Assignment

1. Aim

To preprocess a customer dataset by handling missing values and encoding categorical variables so that the data is suitable for machine learning.

2. Original Dataset

Country	Age	Salary	Purchased
France	44	72000	No
Spain	27	48000	Yes
Germany	30	54000	No
Spain	38	61000	No
Germany	40	Missing	Yes
France	35	58000	Yes
Spain	Missing	52000	No
France	48	79000	Yes
Germany	50	83000	No
France	37	67000	Yes

3. Preprocessing Steps

- Missing Age → mean value = 38.78
- Missing Salary → mean value = 63,777.78
- Country → One-Hot Encoding
- Purchased → No = 0, Yes = 1

4. Python Code

```
import pandas as pd
from sklearn.impute import SimpleImputer

data = {
    "Country": ["France", "Spain", "Germany", "Spain", "Germany",
               "France", "Spain", "France", "Germany", "France"],
    "Age": [44, 27, 30, 38, 40, 35, None, 48, 50, 37],
    "Salary": [72000, 48000, 54000, 61000, None, 58000,
              52000, 79000, 83000, 67000],
    "Purchased": ["No", "Yes", "No", "No", "Yes", "Yes", "No", "Yes", "No", "Yes"]
}

df = pd.DataFrame(data)

# Handle missing values
imputer = SimpleImputer(strategy="mean")
df[["Age", "Salary"]] = imputer.fit_transform(df[["Age", "Salary"]])

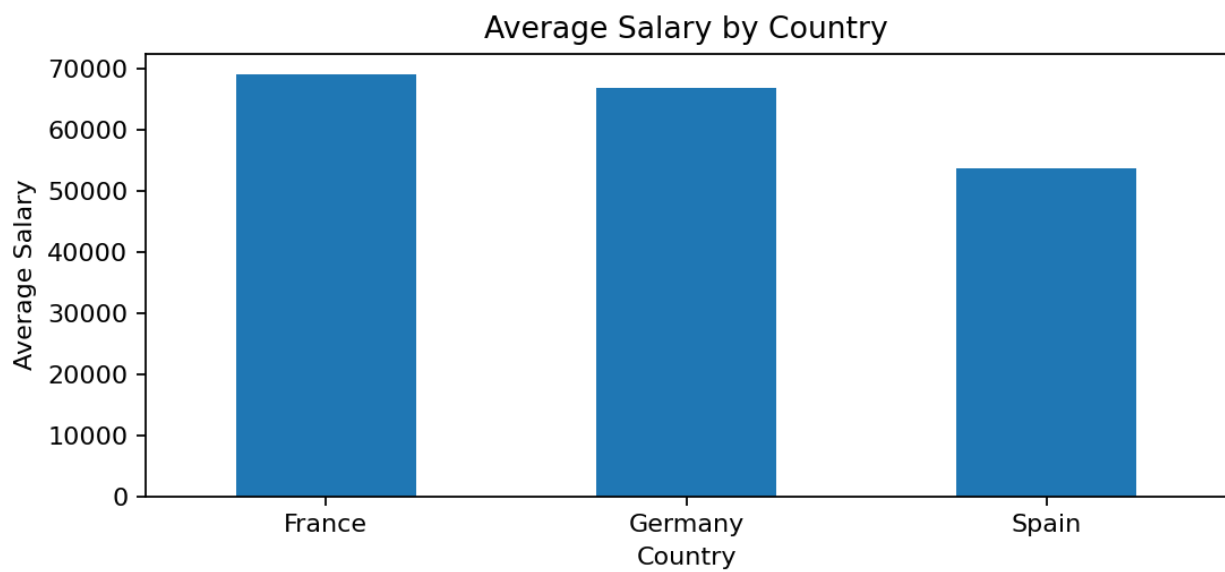
# Encode Purchased
df["Purchased"] = df["Purchased"].map({"No": 0, "Yes": 1})

# One-Hot Encode Country
df = pd.get_dummies(df, columns=["Country"], dtype=int)
```

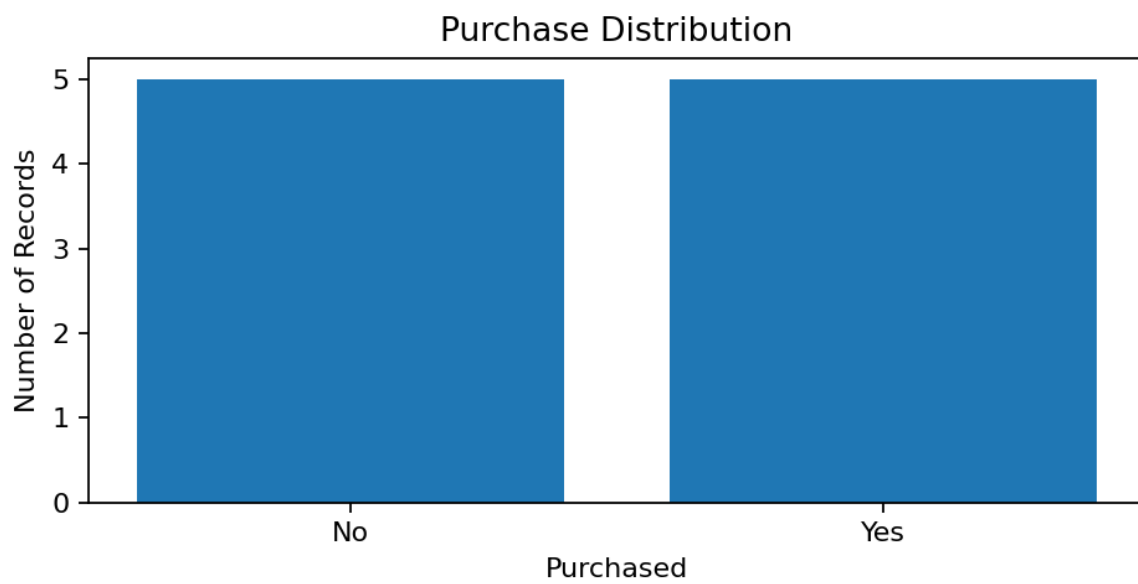
```
print(df)
```

5. Data Visualization

Average salary by country:



Purchase distribution:



6. Observation

- The dataset has 10 records and 4 original attributes.
- Age and Salary contain one missing value each.
- Mean imputation is used for the missing numerical values.
- Country is converted into binary columns using One-Hot Encoding.
- Purchased is converted into 0 and 1.
- The processed dataset contains numerical features suitable for machine learning.

7. Result

The dataset was successfully preprocessed using Python. Missing values were handled, categorical variables were encoded, and the data was converted into a machine-learning-ready format.

8. Conclusion

Data preprocessing improves data quality and converts raw data into a structured format that can be used for machine learning models.